BEBCD
ACDEC
BDABD
BBABE
ABCDD
CBABA

6.2 externalities

Levels of Cleanup	Total Cost of Cleanup ($)	Total Benefit of Cleanup ($)
0	0	0
1	7	45
2	37	80
3	92	105
4	172	125
5	272	140

The table above shows the total cost and the total benefit of cleaning up pollution in a community.

1. Which of the following cleanup levels is socially optimal?

(A) 1

(B) 2 ✓

(C) 3

(D) 4

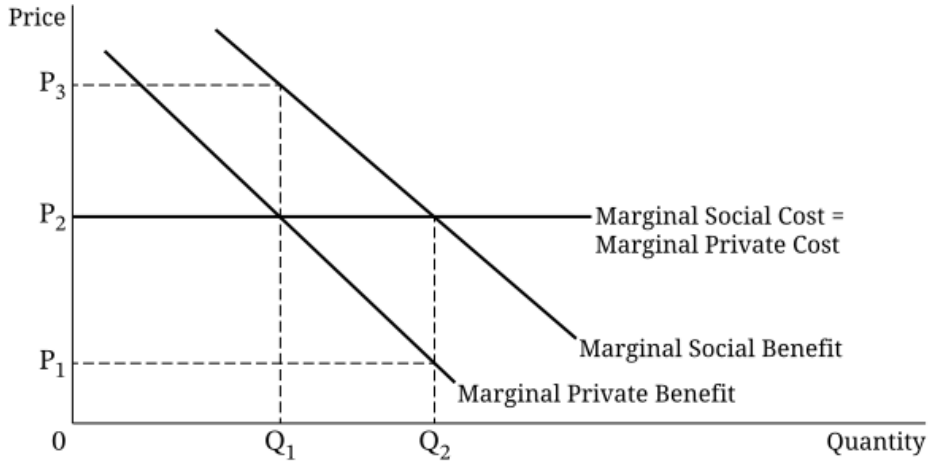
(E) 5

Answer B

Correct. The total cost is rising with cleaning up pollution, while the total benefit is decreasing. Clean up efforts should be undertaken as long as the marginal benefit of additional cleanup exceeds the marginal cost. The optimal level of cleanup is 2 because the marginal benefit of the second cleanup ($35) exceeds the marginal cost ($30). Beginning with the third level of cleanup, the marginal cost ($55) exceeds the marginal benefit ($25). Thus, the optimal level is 2.

6.2 externalities

The graph shows the market equilibrium for a good.



2. What is the socially optimal quantity, and what is the marginal external benefit at that quantity?

(A) Q_1, and the marginal external benefit is $P_3 - P_2$.
(B) Q_1, and the marginal external benefit is $P_2 - P_1$.
(C) Q_2, and the marginal external benefit is P_3.
(D) Q_2, and the marginal external benefit is P_1.
(E) Q_2, and the marginal external benefit is $P_2 - P_1$. ✓

Answer E

Correct. The socially optimal quantity occurs where the marginal social benefit (MSB) equals the marginal social cost (MSC); therefore, the socially optimal quantity is Q_2. The marginal external benefit is given by the difference between the marginal social benefit (MSB) and the marginal private benefit (MPB) curves; therefore, the marginal external benefit is $P_2 - P_1$ at Q_2.

---

3. An industry will produce more than the socially efficient level of output under which of the following conditions?

(A) The production or consumption of a good generates a positive externality.
(B) The production or consumption of a good generates a negative externality. ✓
(C) The industry is a monopoly.
(D) The industry produces a public good.
(E) The industry produces a private good.

6.2 externalities

4. A perfectly competitive manufacturing industry pollutes public water in its production process, leaving the water unsuitable for use by the surrounding communities. At the market equilibrium output level, which of the following is true?

(A) Marginal private cost exceeds marginal social cost.
(B) Marginal private cost exceeds marginal private benefit.
(C) Marginal social cost exceeds marginal social benefit. ✓
(D) The market equilibrium output is equal to the socially efficient output.
(E) The market equilibrium output is less than the socially efficient output.

5. Apple trees need bees to pollinate their flowers and produce apples. Apple farmers hire beekeepers to bring bees to their orchards to pollinate the flowers. This is an example of a

(A) negative externality
(B) tragedy of the commons
(C) public good
(D) market transaction ✓
(E) free-rider problem

Answer D

Correct. A market transaction is a transaction that involves a buyer and a seller. Farmers hiring beekeepers is an example of a market transaction, in which farmers represent the buyers' side of the market and beekeepers represent the sellers' side of the market.

6. Assume that the firms in an industry pollute a river. If there is no government intervention, the firms will

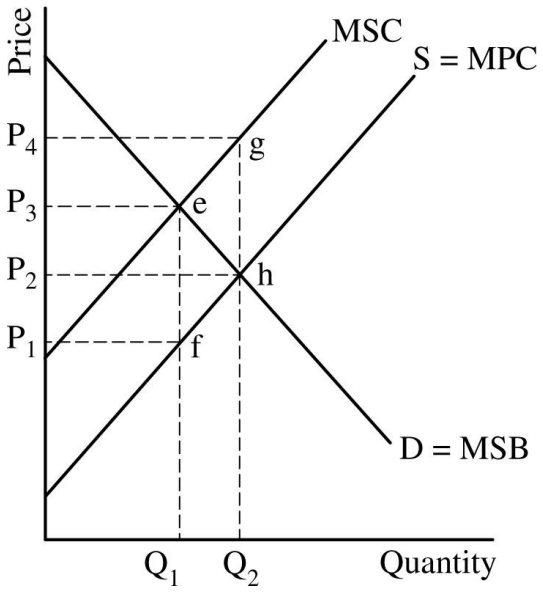
(A) produce more output than is socially efficient ✓
(B) pay production costs that are higher than actual social costs
(C) charge a lower price than is necessary to maximize profit
(D) charge a higher price to pay for the pollution
(E) increase production to achieve allocative efficiency

7. To correct for positive externalities, the government should

6.2 externalities

(A) do nothing, since no harm is done by positive externalities
(B) levy a tax on the output of the good or service
(C) pay a subsidy equal to the marginal external benefit ✓
(D) impose a price ceiling on the good to discourage its production
(E) impose a price floor on the good at which the marginal private benefit equals the marginal social cost

The following questions refer to the following diagram, which shows the marginal social cost (MSC), the marginal private cost (MPC), and the marginal social benefit (MSB) associated with the production of a good in a market.



8. Which of the following describes the type of externality generated by the unregulated private market and the resulting deadweight loss?

6.2 externalities

(A)

Type of Externality	Deadweight Loss
Positive	egh

(B)

Type of Externality	Deadweight Loss
Positive	ehf

(C)

Type of Externality	Deadweight Loss
Negative	ehf

(D) ✓

Type of Externality	Deadweight Loss
Negative	egh

(E)

Type of Externality	Deadweight Loss
Negative	P_4ghP_2

---

9. The use of environmentally friendly drinking straws has benefits to society beyond those enjoyed by their direct users. As a result, a government with the goal of maximizing total economic surplus in society would have an incentive to

(A) allow the free market to determine the equilibrium price and quantity of environmentally friendly drinking straws

(B) implement a binding price ceiling on environmentally friendly drinking straws

(C) implement a binding price floor on environmentally friendly drinking straws

(D) implement a per-unit tax on consumers of environmentally friendly drinking straws

(E) implement a per-unit subsidy to consumers of environmentally friendly drinking straws ✓

6.2 externalities

Answer E

Correct. Since the use of environmentally friendly straws benefits not only the individual but society as well, they generate a positive externality in consumption. In cases where positive externalities are generated in consumption, the good is underconsumed in private markets from the standpoint of maximizing total economic surplus; thus, implementing a per-unit subsidy to consumers would increase consumption towards the point where marginal social benefit equals marginal social cost, therefore increasing total economic surplus towards its maximum.

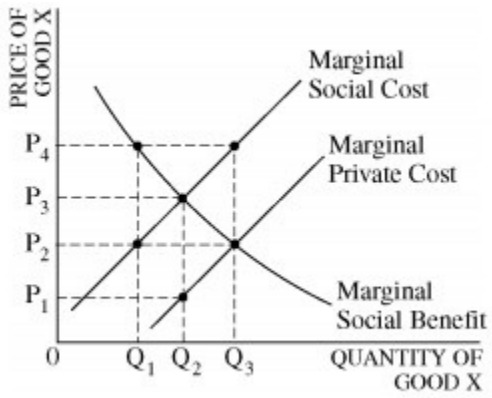
10. Which of the following is always true about a negative externality?

(A) It can be corrected by the government subsidizing the good or activity that creates the externality.
(B) It results in a market output that is smaller than the socially optimal output.
(C) It results in a market output that is greater than the socially optimal output. ✓
(D) It results in a larger deadweight loss than a positive externality does.
(E) It results in a smaller deadweight loss than a positive externality does.

Answer C

Correct. With a negative externality, the marginal social cost is greater than the marginal private cost. The unregulated market output is greater than the socially optimal level of output.

The following questions refer to the diagram below, which shows a market's cost and revenue curves.



11. Given the position of the marginal social cost curve, one can conclude that the

6.2 externalities

(A) private cost of producing good X exceeds the social cost of production at all levels of output

(B) production of good X creates a negative externality ✓

(C) production of good X occurs in an imperfectly competitive market

(D) market quantity, Q_3, is socially optimal quantity

(E) free market will produce too little of good X

12. The socially optimal quantity and the per-unit tax that will achieve the socially optimal quantity are which of the following?

(A)

Socially Optimal Quantity	Per-Unit Tax
Q_1	$P_4 - P_2$

(B)

Socially Optimal Quantity	Per-Unit Tax
Q_2	$P_3 - P_2$

(C)

Socially Optimal Quantity	Per-Unit Tax
Q_3	$P_4 - P_2$

(D) ✓

Socially Optimal Quantity	Per-Unit Tax
Q_2	$P_3 - P_1$

(E)

Socially Optimal Quantity	Per-Unit Tax
Q_3	$P_3 - P_2$

6.2 externalities

13. Good X is produced in a perfectly competitive industry in long-run equilibrium. If the consumption of good X generates positive externalities, the private market will produce

(A) less than the socially optimum amount of good X ✓

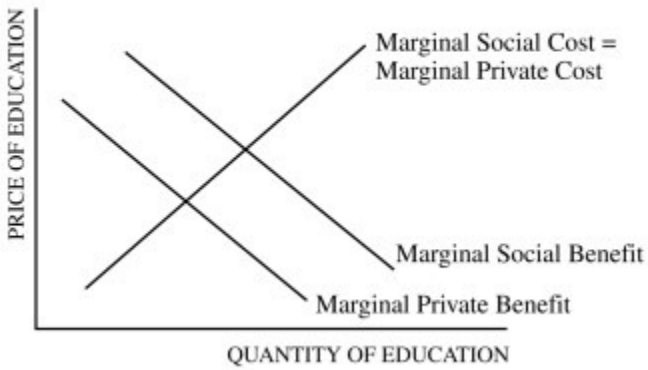
(B) more than the socially optimum amount of good X

(C) the socially optimum amount of good X

(D) good X only if the government grants a subsidy for the production of good X

(E) good X only if the government levies a corrective tax on the production of good X

14.



Which of the following is true based on the graph above?

(A) Quantity of education in the private market is more than the socially optimal quantity.

(B) Consumers of education could be subsidized in order to achieve the socially optimal quantity. ✓

(C) Producers of education could be taxed in order to achieve the socially optimal quantity.

(D) Consumers of education could be taxed in order to achieve the socially optimal quantity.

(E) Consumption of education has a negative externality.

6.2 externalities

Quantity of Pollution Reduction (in tons)	Total Benefit (in dollars)
0	0
1	300
2	550
3	650
4	725
5	775

The table above shows the total benefit associated with reducing different levels of pollution on a small island. The cost of eliminating one ton of pollution is $150.

15. Which of the following is the optimal quantity of pollution to eliminate?

(A) 5 tons
(B) 4 tons
(C) 3 tons
(D) 2 tons ✓
(E) 1 tons

16. If an industry ignores the external costs it generates in its production, which of the following will be true at the competitive market equilibrium output?

(A) Price will be greater than the marginal social cost.
(B) Price will be less than the marginal social cost. ✓
(C) Price will be equal to the marginal social cost.
(D) Marginal private cost will be equal to marginal social cost.
(E) Marginal private cost will be greater than the marginal social cost.

17. If positive externalities exist in the market for flu shots, which of the following is true?

6.2 externalities

(A) Taxing flu shots will lead to the socially efficient level of output.

(B) Subsidizing flu shots will lead to the socially efficient level of output. ✓

(C) The firm making the flu shots is charging too low a price to reach the socially efficient level of output.

(D) The firm making the flu shots is producing where marginal social cost equals marginal social benefit.

(E) The firm making the flu shots is producing the socially efficient quantity if the market for flu shots is perfectly competitive.

18. If the production of a good generates a negative externality, which of the following is true at the private market equilibrium?

(A) The private market equilibrium quantity is greater than the socially optimal quantity. ✓

(B) The private market equilibrium quantity is equal to the socially optimal quantity.

(C) The marginal private cost is greater than the marginal social cost.

(D) The marginal social benefit is greater than the marginal social cost.

(E) The price of the product equals the marginal social cost.

19. If the production of a good generates a positive externality, the government can increase allocative efficiency by

(A) taxing the producer of the good

(B) subsidizing the producer of the good ✓

(C) prosecuting firms that produce the good without proper permits

(D) setting a price ceiling to encourage production of the good

(E) setting a price floor to discourage production of the good

20. If the production of a good results in a positive externality, the government might be able to improve economic efficiency in this market by

(A) eliminating private production of the good

(B) imposing a tax on private producers

(C) promoting the export of the surplus output

(D) initiating antitrust action

(E) granting a subsidy to private producers ✓

21. If the production of good X creates a positive externality, then from society's perspective, the industry that produces good X tends to produce

(A) less than the socially optimal amount of good X ✓

(B) greater than the socially optimal amount of good X

(C) the socially optimal amount of good X

(D) a surplus of good X, unless subsidized

(E) a shortage of good X, unless taxed

6.2 externalities

22. Imposing taxes that increase as a firm's pollution increases is often recommended by economists as a means to reduce pollution. The reason for this recommendation is that such taxes would likely

(A) eliminate pollution completely

(B) encourage firms to use the most efficient method to reduce pollution ✓

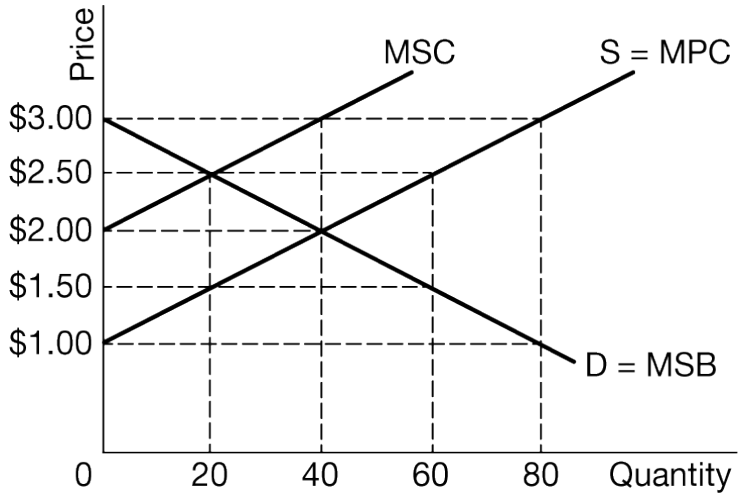
(C) increase the government's revenues

(D) encourage firms to increase production

(E) be paid out of firm's profits and not paid for by higher consumer prices

---

The following graph shows the marginal social cost (MSC), the marginal private cost (MPC), and the marginal social benefit (MSB) of a good.



23. If the government wants the firm to internalize the externality, the government can do so by

(A) doing nothing

(B) imposing a per-unit tax of $0.50

(C) imposing a per-unit tax of $1.00 ✓

(D) granting a per-unit subsidy of $0.50

(E) granting a per-unit subsidy of $1.50

24. Which of the following indicate the marginal external cost and the socially optimal quantity?

(A) $3.00 and 40 units

(B) $2.50 and 60 units

(C) $2.00 and 40 units

(D) $1.00 and 20 units ✓

(E) $1.00 and 80 units

6.2 externalities

Answer D

Correct. The marginal external cost is the vertical distance between the MSC and the MPC at each quantity of output, which is equal to $1.00. The socially optimal quantity occurs where the MSC equals the MSB, which is 20 units.

---

25. Which of the following is NOT an effective way to manage the inefficiency resulting from a negative externality?

(A) Government intervention in the form of obligatory controls

(B) Per-unit taxes

(C) Assignment of property rights

(D) Lump-sum taxes ✓

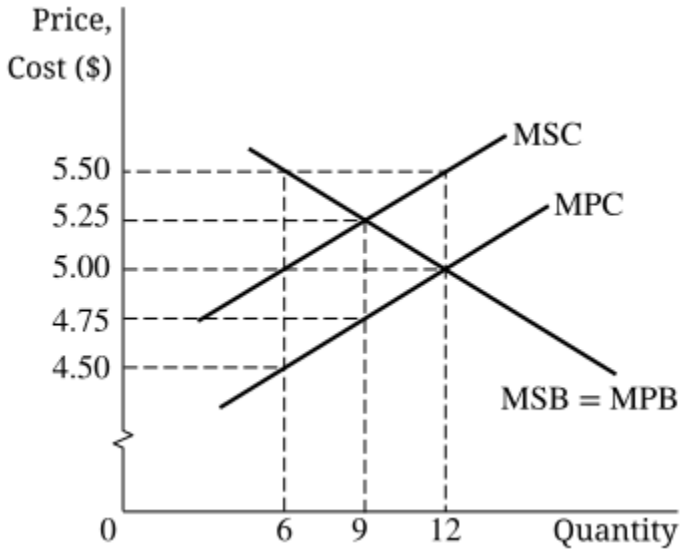
(E) Private negotiations among affected parties

Answer D

Correct. The tax on the negative externality needs to be a per-unit tax based on the amount of the effect on the third person. A lump-sum tax would not be effective because it only affects the fixed cost, not the marginal cost, and therefore it does not affect the profit-maximizing quantity.

6.2 externalities

The following questions are based on the graph provided, which shows the marginal private cost (MPC), marginal social cost (MSC), marginal private benefit (MPB), and marginal social benefit (MSB) in a market for a good that has an external cost.



26. What are the socially efficient and market equilibrium quantities?

(A) Socially efficient quantity is 6, and market equilibrium quantity is 12.

(B) Socially efficient quantity is 9, and market equilibrium quantity is 9.

(C) Socially efficient quantity is 9, and market equilibrium quantity is 12. ✓

(D) Socially efficient quantity is 12, and market equilibrium quantity is 12.

(E) Socially efficient quantity is 12, and market equilibrium quantity is 9.

Answer C

Correct. The socially efficient quantity occurs where marginal social cost equals marginal social benefit (when the quantity is 9 units). The market equilibrium quantity occurs where marginal private cost equals marginal private benefit (when the quantity is 12 units).

27. What is the dollar value of the marginal external cost?

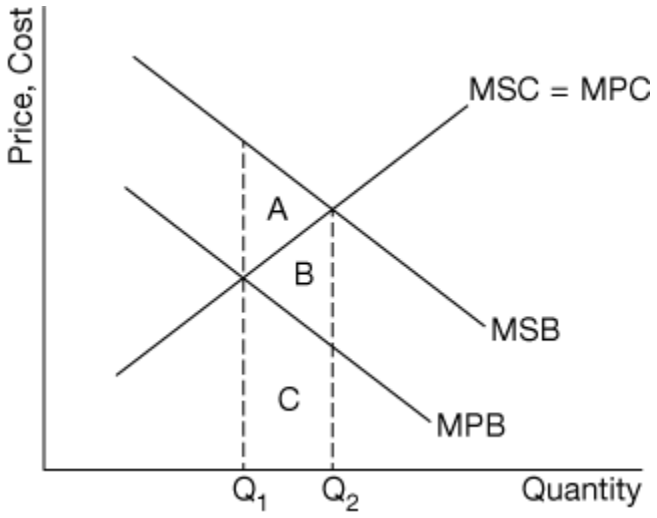
6.2 externalities

(A) $0.25
(B) $0.50 ✓
(C) $1.00
(D) $5.00
(E) $5.50

Answer B

Correct. Marginal external cost is equal to the vertical distance between marginal social cost and marginal private cost at any quantity of output. At the socially optimal quantity of output (9 units), marginal social cost is $5.25 and marginal private cost is $4.75. Therefore, the marginal external cost is $0.50.

28.



The diagram provided illustrates the marginal private benefit (MPB), marginal social benefit (MSB), marginal private cost (MPC), and marginal social cost (MSC) measured in dollars for a competitive market.

Assume that the goal of the government is to increase efficiency in the market. The cost of a government intervention that increases market output from Q_1 to Q_2 is $1.2 million. The intervention will definitely improve market efficiency if the area of

6.2 externalities

(A) A is greater than $\$1.2$ million ✓
(B) B is less than $\$1.2$ million
(C) C is less than $\$1.2$ million
(D) $A+B$ is greater than or equal to $\$1.2$ million
(E) $B+C$ is greater than or equal to $\$1.2$ million

Answer A

Correct. This graph shows a positive externality, where Q_2 is the socially optimal quantity and the deadweight loss is area A. If the cost of the government intervention to increase market output to Q_2, the socially optimum quantity, and eliminates the deadweight loss is $\$1.2$ million, then area A should be greater than or equal to $\$1.2$ million for the intervention to improve market efficiency.

29. Which of the following is true when there are negative externalities associated with the production of a good?

(A) The market will adjust automatically to equate marginal social costs and marginal social benefits.
(B) Marginal social costs will exceed marginal private costs unless businesses are forced to internalize the external costs. ✓
(C) Marginal private costs will exceed marginal social costs, but the government can correct the problem.
(D) Producers should be subsidized so that they will produce more of the good.
(E) Consumers should be subsidized so that they will consume more of the good.

30. There are negative externalities associated with the use of a freeway in a major city at rush hour because during this time

(A) drivers slow down other drivers because of the high traffic volume ✓
(B) drivers value their time more
(C) government revenues from toll roads increase
(D) revenues of bus companies increase
(E) gasoline costs more

31. Which of the following is true if the production of a good creates negative externalities?

(A) The market produces the socially efficient level of output.
(B) Antitrust laws are violated.
(C) The market price equals the marginal social cost.
(D) The marginal social cost exceeds the marginal private cost. ✓
(E) The good is produced by the public sector.

6.2 externalities

Answer D

Correct. When the production of a good generates negative externalities, producers only consider their marginal private cost in determining the level of output to produce. The negative externalities represent the additional (spillover) cost to society. Since marginal social cost accounts for both marginal private cost and marginal external cost (the impact on society because of the negative externality), marginal social costs will always exceed marginal private cost when there are negative externalities in production.

32. Which of the following is true when the production of a good results in negative externalities?

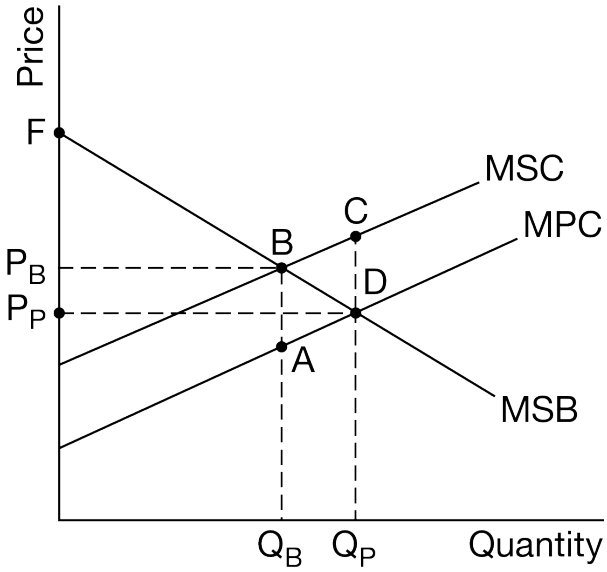
(A) The government must produce the good.
(B) The private market will produce too little of the good.
(C) The private market price will be too low. ✓
(D) The government must prevent the production of the good.
(E) Private firms will not be able to maximize profits.

33. Whenever the production of a good creates negative externalities, an unregulated market will result in

(A) too little of the good being produced
(B) an optimal amount of the good being produced as long as the market is perfectly competitive
(C) society's marginal cost being higher than the firm's marginal cost ✓
(D) the firm's marginal cost being higher than society's marginal cost
(E) the firm's marginal cost being equal to society's marginal cost

6.2 externalities

34.



The graph above depicts the marginal social cost (MSC), marginal private cost (MPC), and marginal social benefit (MSB) associated with the production of a good that generates an externality. Based on the information in the graph, which of the following is true?

(A) There is a positive externality because, relative to the socially optimal output, too little is produced in an unregulated market.

(B) The unregulated market produces $\mathrm{Q_B}$ and sets price at $\mathrm{P_B}$.

(C) The allocatively efficient level of output is $\mathrm{Q_P}$.

(D) The good generates an external benefit equal to AB.

(E) There is a negative externality because marginal social cost exceeds marginal private cost. ✓

Answer E

Correct. This graph illustrates a negative externality because marginal social cost (MSC) exceeds marginal private cost (MPC). The unregulated market will produce too much output (at $\mathrm{Q_P}$) at too low a price relative to the socially optimal output, resulting in an allocative inefficiency. A unit tax or other means to reduce output to $\mathrm{Q_B}$ can be used to correct for the externality and lead the market to produce the allocatively efficient output.

35. Assume the demand curve for a good is perfectly inelastic and the production of each unit of this good generates external costs. A profit-maximizing firm producing the good in an unregulated free market will

6.2 externalities

(A) generate deadweight loss because marginal social cost is greater than marginal private cost
(B) generate deadweight loss only if marginal costs are constant
(C) not generate deadweight loss because the equilibrium quantity is socially optimal ✓
(D) not generate deadweight loss unless marginal costs are constant
(E) not generate deadweight loss unless fixed costs are zero

Answer C

Correct. The demand curve for the good is completely vertical at the market equilibrium quantity. Even though the production of the good entails an external cost (negative externality), the equilibrium quantity produced is the same as the socially optimal quantity; therefore, there would be no deadweight loss.

36. Which of the following is true if the production of a good imposes external costs?

(A) The market supply curve lies above the social supply curve.
(B) An appropriate government policy would be to subsidize production of the good.
(C) A competitive market would not provide the good at all.
(D) A competitive market would produce more than the socially optimal amount. ✓
(E) A persistent market surplus would occur.

37. Which of the following is an example of a negative externality in production?

(A) An electronics store decreases the price of televisions during the holidays.
(B) A car insurance company charges a higher price for teenage drivers.
(C) A local paper mill emits an odor, imposing additional costs on nearby residents. ✓
(D) A pine tree in a public park provides pleasant shade for residents.
(E) A photographer accepts a price for work below marginal cost.

Answer C

Correct. Negative externalities in production occur when the production of a good causes a harmful effect on others not involved in the transaction. A local paper mill emitting an odor imposes additional costs, in the form of a marginal external cost, on nearby residents. Marginal social cost will exceed marginal private cost in this instance; this represents a negative externality in production.

6.2 externalities

38. Which of the following best represents a positive externality?

(A) Purchasing a pass for an amusement park

(B) Dining at a favorite restaurant

(C) Being disturbed by neighboring construction noise

(D) Enjoying watching birds at a neighbor's bird feeder ✓

(E) Dumping waste on someone else's property

39. Which of the following is the best example of a negative externality?

(A) An increase in the price of oil due to the imposition of environmental regulations

(B) An increase in the price of oil due to action taken by the Organization of Petroleum Exporting Countries (OPEC)

(C) A decline in oil stock prices as a result of bad management

(D) Oil leakages from drilling platforms in the Gulf of Mexico ✓

(E) Declining restrictions on the importation of foreign-made cars.

40. Under an efficient policy to address the problem of pollution created from manufacturing good X, which of the following is true?

(A) Pollution is completely eliminated.

(B) The marginal private cost of producing good X exceeds the marginal revenue from its sale.

(C) The marginal private cost of producing good X equals the marginal revenue from its sale.

(D) The marginal social benefit of pollution reduction exceeds the marginal social cost of pollution reduction.

(E) The marginal social benefit of pollution reduction equals the marginal social cost of pollution reduction. ✓

41. In a competitive market in which the production of a good causes pollution, the socially optimal output is different from the competitive market equilibrium output of that good because the

(A) marginal social benefit is higher than the marginal social cost

(B) marginal social benefit is lower than the marginal private cost

(C) marginal social cost is higher than the marginal private cost ✓

(D) marginal social benefit is higher than the marginal private benefit

(E) total social cost is less than the total social benefit

Answer C

Correct. In the presence of negative externalities (e.g., pollution), the marginal social cost exceeds the marginal private cost, and the marginal social benefit is less than the marginal social cost.

6.2 externalities

42. If a positive externality exists in the consumption of a good, the private market equilibrium quantity will be

(A) the same as the socially optimal quantity, since the marginal private benefit equals the marginal social benefit

(B) greater than the socially optimal quantity, since the marginal social cost exceeds the marginal private cost

(C) greater than the socially optimal quantity, since the marginal private benefit exceeds the marginal social benefit

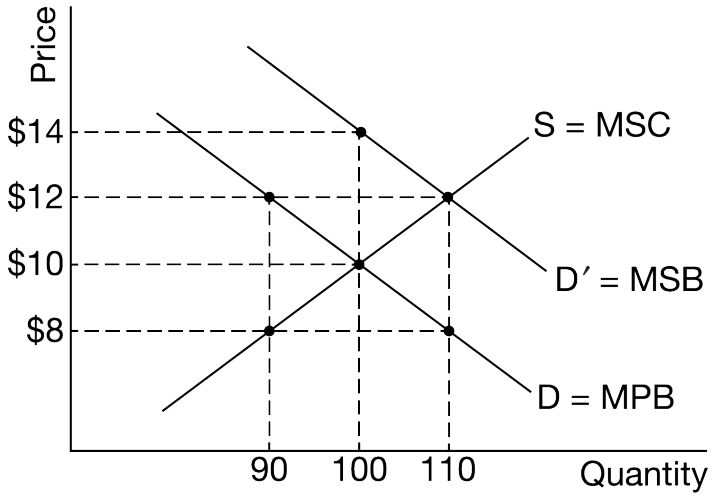
(D) less than the socially optimal quantity, since the marginal social benefit exceeds the marginal private benefit ✓

(E) less than the socially optimal quantity, since the marginal private benefit exceeds the marginal private cost

Answer D

Correct. The unregulated market will not internalize the externality, since private agents only consider marginal private benefit and marginal private cost when making decisions. With a positive externality, the marginal social benefit (MSB) exceeds the marginal private benefit (MPB). The private market equilibrium quantity will be less than the socially optimal quantity.

43.



The graph above depicts the marginal private benefit (MPB), the marginal social benefit (MSB), and the marginal social cost (MSC) of producing a good. Which of the following correctly identifies both the problem illustrated in the graph and the remedy to the problem?

6.2 externalities

(A) Negative externality; a corrective per-unit tax equal to the external cost of $2 will lead to the socially optimal output.

(B) Negative externality; a corrective per-unit tax equal to the external cost of $4 will lead to the socially optimal output.

(C) Positive externality; a corrective per-unit subsidy equal to the external benefit of $2 will lead to the socially optimal output.

(D) Positive externality; a corrective per-unit subsidy equal to the external benefit of $4 will lead to the socially optimal output. ✓

(E) Positive externality; a corrective per-unit tax equal to the external benefit of $6 will lead to the socially optimal output.

Answer D

Correct. The graph illustrates a positive externality because the $\text{MSB} > \text{MPB}$ at all levels of output. At the market equilibrium, there is a marginal external benefit equal to $4 (not $2), which is equal to the vertical distance between MSB and MPB at each output level. That is, $(\$12 - \$8), or (\$14 - \$10) = \$4.$ Therefore, a $4 per-unit subsidy would correct for the externality.

44. Which of the following is an example of a positive externality?

(A) Traffic congestion in cities

(B) Emission controls on power plants

(C) Immunizations that prevent the spread of diseases ✓

(D) Tariffs that protect a domestic industry

(E) Secondhand smoke from cigarettes

Answer C

Correct. With a positive externality, the marginal social benefit exceeds the marginal private benefit at all levels of output of a good or a service consumed. Being vaccinated against contagious diseases protects not only the individuals vaccinated, but also those who are not vaccinated. Thus, the marginal social benefit of immunizations exceeds the marginal private benefit.

45. When consumption of a good generates a positive externality, which of the following must be true at the market equilibrium?

6.2 externalities

(A) Marginal social benefit is less than marginal private cost.

(B) Marginal social benefit is greater than marginal private benefit. ✓

(C) Marginal social cost is greater than marginal social benefit.

(D) Marginal social cost is less than marginal private benefit.

(E) Marginal social cost is equal to marginal external benefit.

46. Which of the following would create a positive externality?

(A) Congestion on city highways

(B) Water pollution

(C) A neighbor's loud music that wakes you up

(D) Flu vaccination ✓

(E) The greenhouse effect and global warming

47. Which of the following is most likely true if textile production generates pollution during the production process?

(A) The market will underproduce textiles relative to the socially optimal quantity, and the government would need to provide a lump-sum subsidy to firms to correct the externality.

(B) The market will underproduce textiles relative to the socially optimal quantity, and the government would need to provide a per-unit subsidy to firms to correct the externality.

(C) The market will overproduce textiles relative to the socially optimal quantity, and the government would need to impose a lump-sum tax on firms to correct the externality.

(D) The market will overproduce textiles relative to the socially optimal quantity, and the government would need to impose a per-unit tax on firms to correct the externality. ✓

(E) The market will produce the socially optimal quantity of textiles, and the government would not need to intervene in the market to influence the production decisions of firms.

Answer D

Correct. The generation of pollution is a negative externality in production. When negative externalities are generated in production, marginal private cost is lower than marginal social cost; thus, the market will overproduce textiles relative to the socially optimal quantity, since only marginal private cost is considered. To correct for the negative production externality, the government would need to impose a per-unit tax on firms to internalize the externality and consider the full social cost of textile production.